

Courier and Parcel Delivery Management System

Entity Relationship (ER) Diagram & Relational Schema

Prepared by	Application Domain	Submission
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Objective

Design a complete Entity Relationship (ER) Diagram for the application selected in the First DBMS Assignment and convert the design into a relational schema. The design identifies entities, attributes, primary keys, relationships, cardinalities, foreign keys, and the resulting relations.

Application Purpose

The system manages the complete life cycle of a parcel, starting from booking at a branch, assignment to a delivery employee and vehicle, tracking movement between branches, final delivery to the receiver, payment processing, and complaints.

Submission Contents

No.	Section
1	Entities and Attributes
2	Relationships and Cardinalities
3	Complete ER Diagram
4	Relational Schema
5	Primary and Foreign Key Identification
6	Relationship Explanations

Design note: The nine entities and relationship structure are based on the First DBMS Assignment. The ER diagram uses standard Chen-style notation and explicitly shows PK/FK attributes and cardinalities.

1. Entities and Attributes

The system contains nine entities. Each entity represents a distinct type of information required by the courier application.

Entity	Attributes	Primary Key
Customer	CustomerID , Name, Address, City, Phone, Email	CustomerID
Branch	BranchID , BranchName, City, Address, ContactNumber	BranchID
Employee	EmployeeID , Name, Designation, <u>BranchID</u> , Phone, DateOfJoining	EmployeeID
Vehicle	VehicleID , VehicleNumber, VehicleType, <u>BranchID</u>	VehicleID
Parcel	ParcelID , <u>CustomerID</u> , ReceiverName, ReceiverAddress, ReceiverPhone, Weight, ParcelType, <u>BookingBranchID</u> , <u>DestinationBranchID</u> , BookingDate, DeliveryStatus	ParcelID
Delivery_Assignment	AssignmentID , <u>ParcelID</u> , <u>EmployeeID</u> , <u>VehicleID</u> , AssignedDate, ExpectedDeliveryDate, ActualDeliveryDate	AssignmentID
Parcel_Tracking	TrackingID , <u>ParcelID</u> , CurrentLocation, Status, StatusDateTime	TrackingID
Payment	PaymentID , <u>ParcelID</u> , AmountPaid, PaymentMode, PaymentDate, PaymentStatus	PaymentID
Complaint	ComplaintID , <u>ParcelID</u> , <u>CustomerID</u> , ComplaintDate, Description, ComplaintStatus	ComplaintID

Key convention: PK = Primary Key; FK = Foreign Key. In the final ERD, PK attributes are underlined and FK attributes are labelled.

2. Relationships and Cardinalities

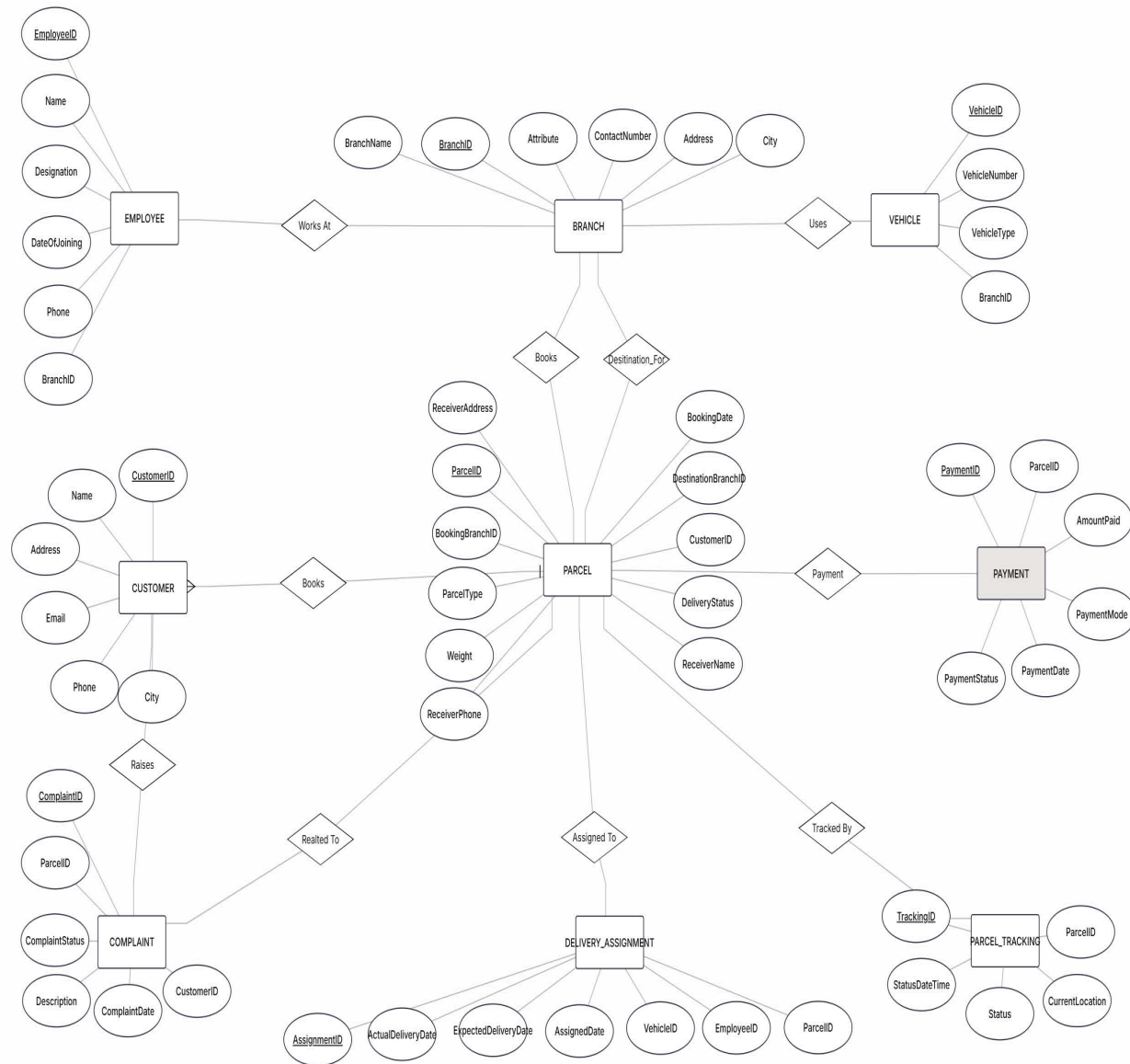
The following relationships are derived from the original application design. Branch–Parcel is represented by two separate relationships because Parcel contains both BookingBranchID and DestinationBranchID.

Entity A	Entity B	Relationship	Cardinality	Explanation
Branch	Employee	Works At / Employs	1 : M	One branch can employ many employees; each employee belongs to one branch.
Branch	Vehicle	Uses / Stations	1 : M	One branch can have many vehicles; each vehicle is stationed at one branch.
Customer	Parcel	Books	1 : M	One customer can book many parcels; each parcel is booked by one customer.
Branch	Parcel	Books (Booking Branch)	1 : M	A branch can be the booking branch for many parcels.
Branch	Parcel	Destination For	1 : M	A branch can be the destination branch for many parcels.
Parcel	Delivery_Assignment	Has Assignment	1 : M	A parcel may have multiple assignments when reassignment/redelivery is required.
Employee	Delivery_Assignment	Handles	1 : M	One employee can handle many delivery assignments over time.
Vehicle	Delivery_Assignment	Used In	1 : M	One vehicle can be used for many delivery assignments over time.
Parcel	Parcel_Tracking	Has Tracking	1 : M	One parcel can have many tracking updates.
Parcel	Payment	Has Payment	1 : 1	Each parcel has one corresponding payment record in this design.
Customer	Complaint	Raises	1 : M	One customer may raise complaints for multiple parcels.
Parcel	Complaint	Has Complaint	1 : M	One parcel may have multiple complaints.

M:N relationship check: The original design does not specify an M:N relationship that requires a new associative table. Delivery_Assignment is already a separate entity connecting Parcel, Employee and Vehicle through 1:M relationships.

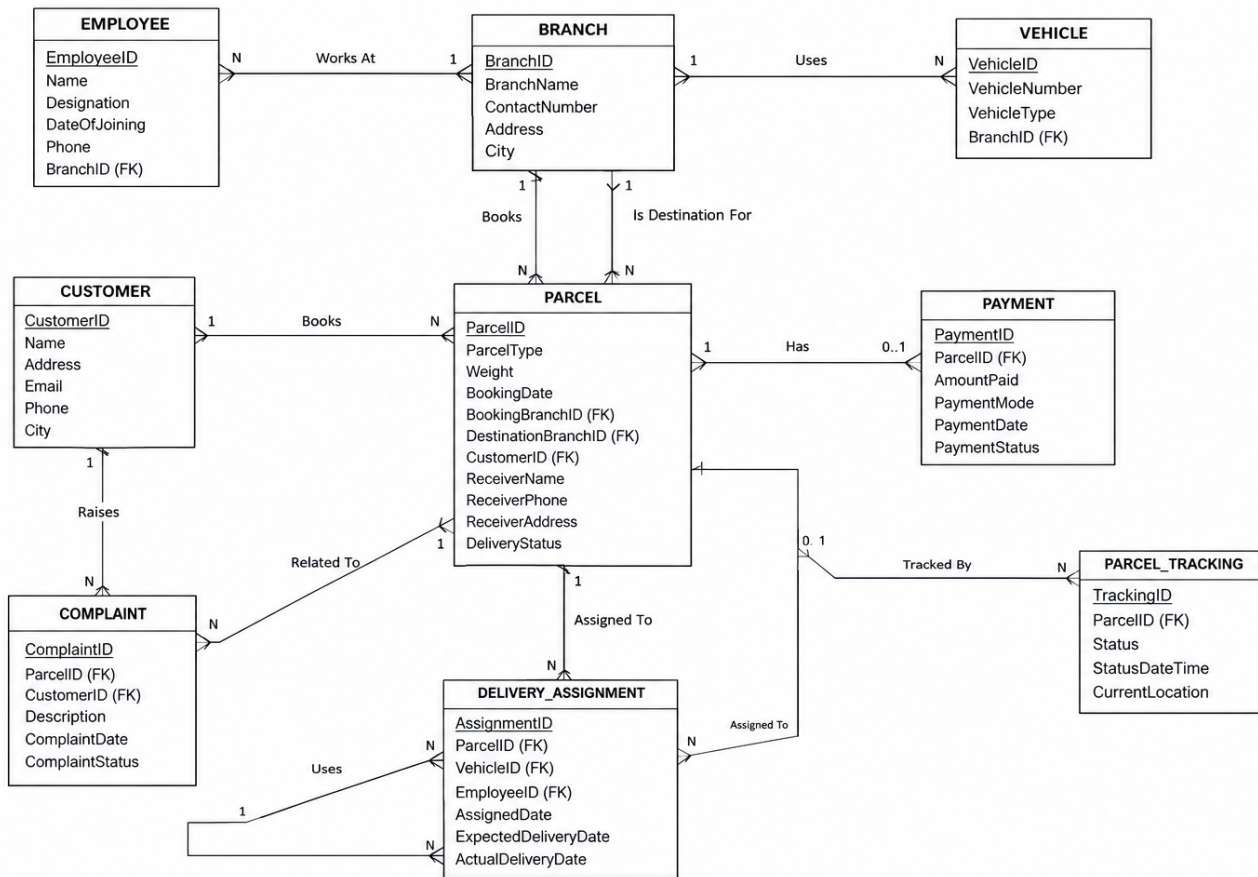
Courier and Parcel Delivery Management System — ER Diagram

ERDPlus Diagram



4. Relational Schema

Each entity is converted into a relation (table). Primary keys uniquely identify rows, while foreign keys establish references to related relations.



5. Primary Key and Foreign Key Identification

The following table provides a direct checklist for implementing the relational schema.

Table	Primary Key (PK)	Foreign Key(s) (FK)
Customer	CustomerID	—
Branch	BranchID	—
Employee	EmployeeID	BranchID → Branch(BranchID)
Vehicle	VehicleID	BranchID → Branch(BranchID)
Parcel	ParcelID	CustomerID → Customer(CustomerID) BookingBranchID → Branch(BranchID) DestinationBranchID → Branch(BranchID)
Delivery_Assignment	AssignmentID	ParcelID → Parcel(ParcelID) EmployeeID → Employee(EmployeeID) VehicleID → Vehicle(VehicleID)
Parcel_Tracking	TrackingID	ParcelID → Parcel(ParcelID)
Payment	PaymentID	ParcelID → Parcel(ParcelID)
Complaint	ComplaintID	ParcelID → Parcel(ParcelID) CustomerID → Customer(CustomerID)

Foreign-key mapping ensures referential links between the relations. Parcel is the central relation and contains references to Customer and Branch in addition to its own delivery and receiver information.

6. Relationship Explanation

1. Branch — Employee (1 : M)

One branch can employ many employees; each employee belongs to one branch.

2. Branch — Vehicle (1 : M)

One branch can have many vehicles; each vehicle is stationed at one branch.

3. Customer — Parcel (1 : M)

One customer can book many parcels; each parcel is booked by one customer.

4. Branch — Parcel (1 : M)

A branch can be the booking branch for many parcels.

5. Branch — Parcel (1 : M)

A branch can be the destination branch for many parcels.

6. Parcel — Delivery_Assignment (1 : M)

A parcel may have multiple assignments when reassignment/redelivery is required.

7. Employee — Delivery_Assignment (1 : M)

One employee can handle many delivery assignments over time.

8. Vehicle — Delivery_Assignment (1 : M)

One vehicle can be used for many delivery assignments over time.

9. Parcel — Parcel_Tracking (1 : M)

One parcel can have many tracking updates.

10. Parcel — Payment (1 : 1)

Each parcel has one corresponding payment record in this design.

11. Customer — Complaint (1 : M)

One customer may raise complaints for multiple parcels.

12. Parcel — Complaint (1 : M)

One parcel may have multiple complaints.

Conclusion

The Courier and Parcel Delivery Management System is represented by nine interconnected relations: Customer, Branch, Employee, Vehicle, Parcel, Delivery_Assignment, Parcel_Tracking, Payment and Complaint. The ER model captures parcel booking, branch routing, employee and vehicle assignment, tracking, payment and complaint management. The relational schema preserves these connections through primary and foreign keys.

ERDPlus submission note: Recreate the ER diagram page in ERDPlus using the same nine entities, attributes, relationship names and cardinalities shown in this document. Then export the ERDPlus diagram and use it as the final digital ERD required by the assignment.